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Preparation of Luria–Bertani (LB) Agar Plates

1 Purpose

This procedure describes the preparation, sterilisation and pouring of LB agar plates for routine bacterial culture.

2 Scope

It applies to all staff and students who prepare media in the Media Preparation Unit.

3 Safety

- Wear a laboratory coat, safety glasses and heat-resistant gloves when handling hot media.
- Never open the autoclave until the chamber pressure has returned to zero.
- Report any burn or spillage to the unit supervisor immediately.

4 Materials

Component	Amount per litre	Grade
Tryptone	10 g	Microbiology grade
Yeast extract	5 g	Microbiology grade
Sodium chloride	10 g	Analytical reagent
Agar	15 g	Bacteriological
Purified water	to 1000 mL	Type II

5 Procedure

1. Weigh the dry components into a 2 L glass bottle.
2. Add 900 mL of purified water and stir until the powders are fully suspended.
3. Adjust the volume to 1000 mL.
4. Sterilise by autoclaving:
    a. 121 °C for 15 minutes on the liquid cycle;
    b. loosen the cap by a quarter turn before loading.
5. Cool the medium to about 50 °C in a water bath.
6. If required, add the antibiotic aseptically and swirl gently to mix.
7. Pour approximately 20 mL into each sterile Petri dish and allow the plates to set.

6 Storage and records

Store the plates inverted at 2–8 °C and use them within four weeks. Record the batch number, date and preparer in the media log. Discard any plate that shows contamination, cracking or drying.